Chapter 17, Problem 44P

Problem

Design a problem to help other students better understand how to find the rms voltage across and the rms current through an electrical element given a Fourier series for both the current and the voltage. In addition, have them calculate the average power delivered to the element and the power spectrum.

Step-by-step solution

Step 1 of 4

Let us consider the Fourier series for both current and voltage are,

$$v(t) = V_{dc} + V_a \cos(n_1\omega_0 t - \alpha_n) + V_b \cos(n_2\omega_0 t - \alpha_k) \text{ V}$$

$$i(t) = I_{dc} + I_a \cos(m_1\omega_0 t - \beta_m) \text{ A}$$

The rms value of the voltage

$$V_{\text{rms}} = \sqrt{a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}(a_n + b_n)^2}$$

$$\therefore \boxed{V_{\text{rms}} = \sqrt{V_{dc}^2 + \frac{1}{2}\left[V_a^2 + V_b^2\right]} \text{ V}}$$

Step 2 of 4

The rms value of the current

$$I_{\text{rms}} = \sqrt{a_0 + \frac{1}{2}\sum_{n=1}^{\infty}\left(a_n^2 + b_n^2\right)}$$

$$I_{\text{rms}} = \sqrt{I_{dc}^2 + \frac{1}{2}\left[I_a^2 + 0\right]}$$

$$\therefore \boxed{I_{\text{rms}} = \sqrt{I_{dc}^2 + \frac{I_a^2}{2}} \text{ A}}$$

Step 3 of 4

Average power delivered to the element is,

$$P = V_{dc}I_{dc} + \frac{1}{2}\sum_{n=1}^{\infty}V_n I_n \cos(\theta_n - \phi_n)$$

Where $m = n$

$$P = V_{dc}I_{dc} + \frac{1}{2}\left[(V_a)(I_a)\cos(\alpha_n - \beta_n) + 0\right]$$

$$\therefore \boxed{P = V_{dc}I_{dc} + \frac{V_a I_a}{2}\cos(\alpha_n - \beta_n) \text{ W}}$$

Step 4 of 4

Thus, the power spectrum is